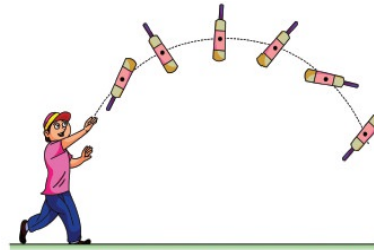


Lecture 7 – Translational motion of rigid bodies

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In this lecture we consider the translational motion of a **rigid body**, which may be considered as being made up of a large number of particles with masses m_i . We will see how the concept of the **center-of-mass** of a rigid body yields a simple understanding of the motion of rigid bodies.



A few slides in this chapter come from Chapter-8 of Young's textbook. This PPT file will well define the scope of your tests.

1

Center-of-mass position

Consider a system of two particles with **equal masses m** , lying on the x -axis:

What is the position for the **center-of-mass** of the system with two particles in each case?

$$x_1 = -2.5 \text{ m}, \quad x_2 = 2.5 \text{ m}$$

$$x_1 = 0, \quad x_2 = 5 \text{ m}$$

$$x_1 = -5 \text{ m}, \quad x_2 = 0$$

$$x_1 = -2 \text{ m}, \quad x_2 = 3 \text{ m}$$

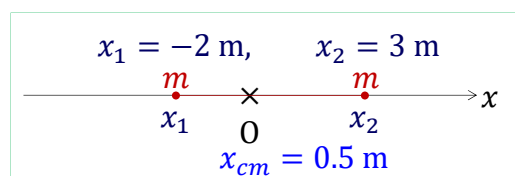
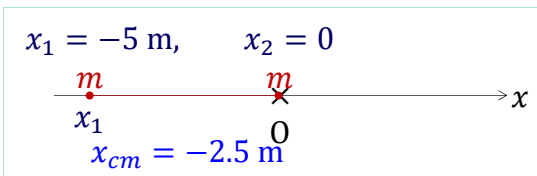
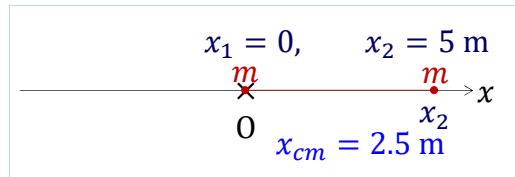
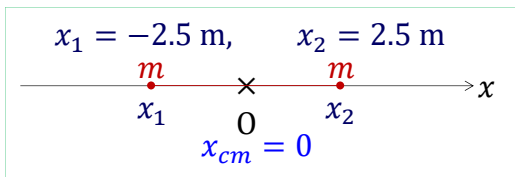
2

Center-of-mass position (cont.)

What is the **center-of-mass position** of the system with two particles in each case?

In these cases we have

$$x_{cm} = \frac{mx_1 + mx_2}{m + m} = \frac{1}{2}(x_1 + x_2).$$

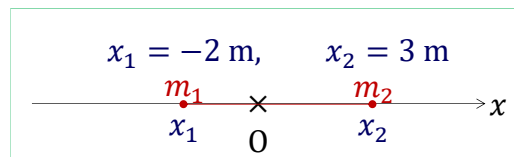
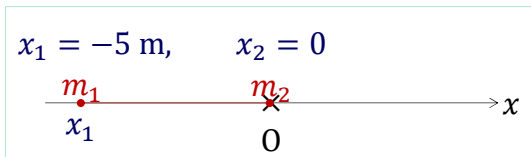
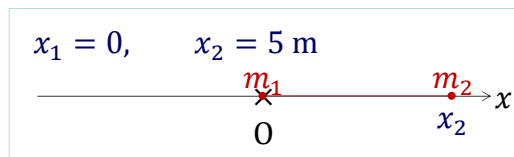
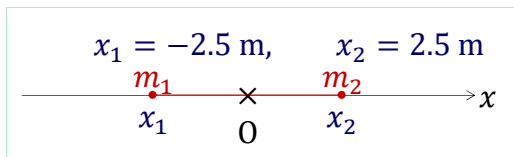


3

Center-of-mass position (cont.)

Now consider a system of two particles with **unequal masses** m_1 and m_2 , lying on the x -axis.

What is the position for the **center-of-mass** of the system with two particles in each case?



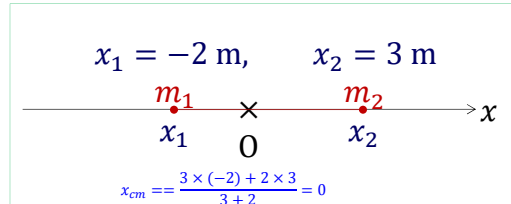
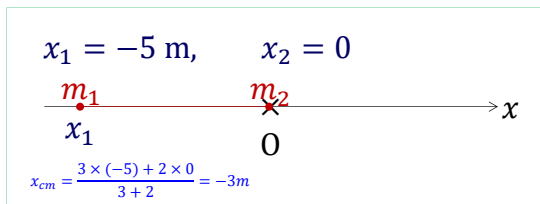
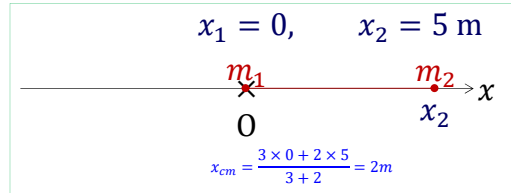
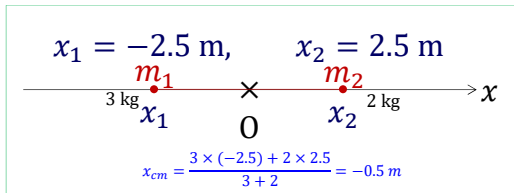
4

Center-of-mass position (cont.)

If $m_1 = 3 \text{ kg}$ and $m_2 = 2 \text{ kg}$, what is the **center-of-mass position** of the system of two particles in each case?

In general, we have

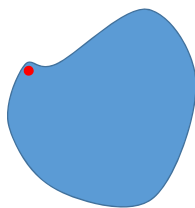
$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}.$$



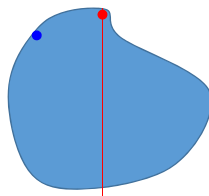
5

Center-of-mass position – center of gravity

Find the center of gravity for this irregular cardboard.

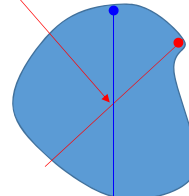


Front View



Front View

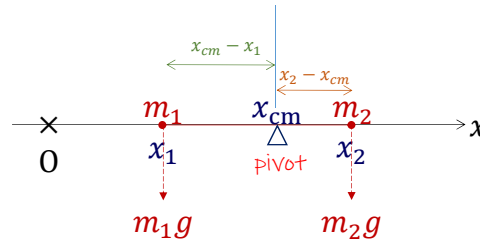
Place a pivot below this point



Plane View

6

Center-of-mass position – center of gravity



The **center of gravity** is such that

$$m_1 g \times (x_{cm} - x_1) = m_2 g \times (x_2 - x_{cm})$$

$$\rightarrow m_1 x_{cm} - m_1 x_1 = m_2 x_2 - m_2 x_{cm}$$

$$\rightarrow (m_1 + m_2) x_{cm} = m_1 x_1 + m_2 x_2$$

$$\rightarrow x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

Taking the moment with respect to the pivot!!

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Center-of-mass position (cont.)

If the system of two particles with masses m_1 and m_2 now lie on the xy -plane, its **center-of-mass position** is given by (x_{cm}, y_{cm}) with

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}.$$

If $(x_{cm}, y_{cm}) = (0, 0)$, x_i and y_i will be the distance measured from the origin.

If the particles are in motion, then

$$x_1 = x_1(t), \quad y_1 = y_1(t); \quad x_2 = x_2(t), \quad y_2 = y_2(t).$$

$$\text{and, } x_{cm}(t) = \frac{m_1 x_1(t) + m_2 x_2(t)}{m_1 + m_2}$$

$$y_{cm}(t) = \frac{m_1 y_1(t) + m_2 y_2(t)}{m_1 + m_2}$$

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Center-of-mass velocity and total linear momentum

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}.$$

The **velocity** at the **center-of-mass** of the system of two particles of masses m_1 and m_2 is given by $(v_{cm,x}, v_{cm,y})$ with

$$v_{cm,x} = \frac{dx_{cm}}{dt} = \frac{m_1}{m_1 + m_2} \frac{dx_1}{dt} + \frac{m_2}{m_1 + m_2} \frac{dx_2}{dt} = \frac{m_1}{m_1 + m_2} v_{1x} + \frac{m_2}{m_1 + m_2} v_{2x}$$

$$v_{cm,y} = \frac{dy_{cm}}{dt} = \frac{m_1}{m_1 + m_2} \frac{dy_1}{dt} + \frac{m_2}{m_1 + m_2} \frac{dy_2}{dt} = \frac{m_1}{m_1 + m_2} v_{1y} + \frac{m_2}{m_1 + m_2} v_{2y}$$

Multiplying through with the total mass of the system of two particles, i.e., $m_1 + m_2 = M$, we have

$$M v_{cm,x} = m_1 v_{1x} + m_2 v_{2x}, \quad M v_{cm,y} = m_1 v_{1y} + m_2 v_{2y}.$$

This is the **total linear momentum** (p_x, p_y) of the system of two particles, with

$$M v_{cm,x} = p_x = p_{1x} + p_{2x}, \quad M v_{cm,y} = p_y = p_{1y} + p_{2y}.$$

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Center-of-mass velocity and total linear momentum (cont.)

If the **system of two particles** is **isolated** and therefore the **total linear momentum** is **conserved**, then

$$p_x = p_{1x} + p_{2x} = \text{constant}, \quad p_y = p_{1y} + p_{2y} = \text{constant},$$

$$\text{or, } M v_{cm,x} = \text{constant}, \quad M v_{cm,y} = \text{constant}.$$

$$\text{or, } v_{cm,x} = \text{constant}, \quad v_{cm,y} = \text{constant}.$$

If these constants are equal to zero, then

$$v_{cm,x} = \frac{dx_{cm}}{dt} = 0, \quad v_{cm,y} = \frac{dy_{cm}}{dt} = 0,$$

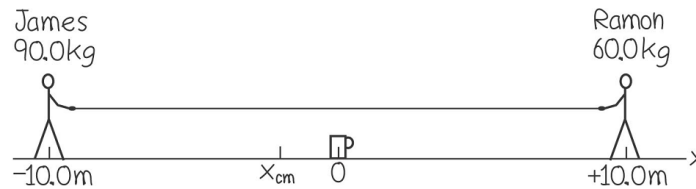
$$\text{or, } x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \text{constant},$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \text{constant}.$$

If the velocity at the center of mass of a system of particles is 0, the coordinate for the center of mass for that system of particles will not change.

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Example 1. An isolated system of two objects with zero total linear momentum



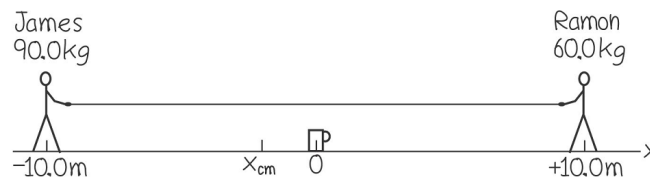
James and Ramon are 20.0 m apart on a frozen pond. Midway between them is a mug of their favourite beverage. Their **centre-of-mass position** is given by

$$x_{cm} = \frac{90}{90 + 60} \times (-10) + \frac{60}{90 + 60} \times 10 = -\frac{300}{150} = -2 \text{ m.}$$

James and Ramon is an example of an **isolated system of two objects**. Their total linear momentum is zero and will remain zero as long as they are isolated. Then, $v_{cm,x} = 0 \Rightarrow x_{cm} = -2 \text{ m is a constant.}$

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Example 1. An isolated system of two objects with zero total linear momentum (cont.)



The objects can move!! But the total linear momentum is still 0.

James and Ramon can move by pulling on the ends of a light rope stretched between them. This does not change the fact that they constitute an isolated system of two objects (how?). So,

$$\frac{90}{90 + 60} x_{\text{James}}^{(-10)} + \frac{60}{90 + 60} x_{\text{Ramon}}^{(10)} = -2$$

Position for the center of mass computed previously

When James has moved 6.00 m toward the mug, his new position is

$$x_{\text{James}} = -4 \text{ m, and } \frac{3}{5} \times (-4) + \frac{2}{5} x_{\text{Ramon}} = \underbrace{x_{cm}}_{\text{no change}} = -2 \rightarrow x_{\text{Ramon}} = 1 \text{ m.}$$

Ramon must at the same time have moved 9.00 m toward the mug.

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Center-of-mass acceleration and Newton's equation of motion

The **acceleration** at the **center-of-mass** of the system of two particles is given by $(a_{cm,x}, a_{cm,y})$ with

$$a_{cm,x} = \frac{dv_{cm,x}}{dt} = \frac{m_1}{m_1 + m_2} \frac{dv_{1x}}{dt} + \frac{m_2}{m_1 + m_2} \frac{dv_{2x}}{dt} = \frac{m_1}{m_1 + m_2} a_{1x} + \frac{m_2}{m_1 + m_2} a_{2x}$$

$$a_{cm,y} = \frac{dv_{cm,y}}{dt} = \frac{m_1}{m_1 + m_2} \frac{dv_{1y}}{dt} + \frac{m_2}{m_1 + m_2} \frac{dv_{2y}}{dt} = \frac{m_1}{m_1 + m_2} a_{1y} + \frac{m_2}{m_1 + m_2} a_{2y}.$$

Multiplying through with the total mass of the system of two particles, i.e., $m_1 + m_2 = M$, we have

$$Ma_{cm,x} = m_1 a_{1x} + m_2 a_{2x}, \quad Ma_{cm,y} = m_1 a_{1y} + m_2 a_{2y}.$$

This is equal to the **total (external) force** (F_x, F_y) acting on the system of two particles, with

$$Ma_{cm,x} = F_x = m_1 a_{1x} + m_2 a_{2x}, \quad Ma_{cm,y} = F_y = m_1 a_{1y} + m_2 a_{2y}.$$

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Center-of-mass acceleration and Newton's equation of motion (cont.)

$$F_x = Ma_{cm,x}, \quad F_y = Ma_{cm,y}$$

(F_x, F_y) is the **total (external) force** acting on the system of two particles, since the **total (internal) force** is equal to zero:

$$F_{1 \text{ on } 2, x} = -F_{2 \text{ on } 1, x}, \quad F_{1 \text{ on } 2, y} = -F_{2 \text{ on } 1, y}.$$

and,

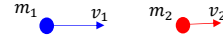
$$F_{1 \text{ on } 2, x} + F_{2 \text{ on } 1, x} = 0, \quad F_{1 \text{ on } 2, y} + F_{2 \text{ on } 1, y} = 0.$$

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Center-of-mass position, velocity, and acceleration – in summary

For a system of two particles with masses m_1 and m_2 moving on the xy -plane, its **center-of-mass position** is given by (x_{cm}, y_{cm}) with

$$x_{cm} = \frac{1}{M}(m_1x_1 + m_2x_2), \quad y_{cm} = \frac{1}{M}(m_1y_1 + m_2y_2).$$



Its **velocity** at the **centre-of-mass** is given by $(v_{cm,x}, v_{cm,y})$ with

$$v_{cm,x} = \frac{1}{M}(m_1v_{1x} + m_2v_{2x}), \quad v_{cm,y} = \frac{1}{M}(m_1v_{1y} + m_2v_{2y}).$$

The **total linear momentum**,

$$p_x = Mv_{cm,x}, \quad p_y = Mv_{cm,y}.$$

Its **acceleration** at the **centre-of-mass** is given by $(a_{cm,x}, a_{cm,y})$ with

$$a_{cm,x} = \frac{1}{M}(m_1a_{1x} + m_2a_{2x}), \quad a_{cm,y} = \frac{1}{M}(m_1a_{1y} + m_2a_{2y}).$$

The **total (external) force** acting on the system, $F_x = Ma_{cm,x}$ and $F_y = Ma_{cm,y}$.

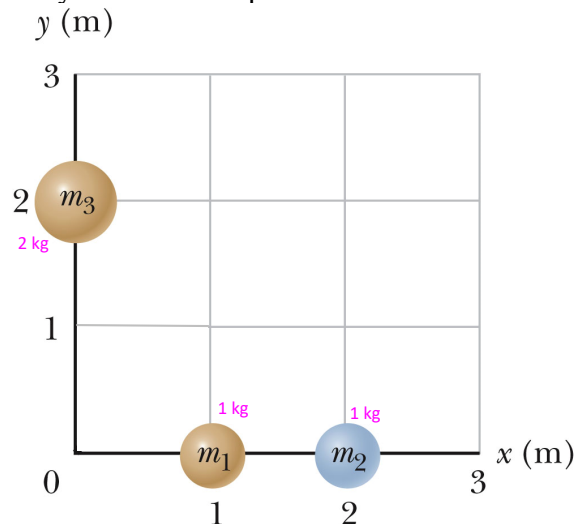
15

Example 2. The center-of-mass position of a system of three particles

A system consists of three particles with masses $m_1 = m_2 = 1.0$ kg and $m_3 = 2.0$ kg located as shown. The **center-of-mass** of the system is given by

$$\begin{aligned} x_{cm} &= \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} \\ &= \frac{(1 + 2 + 0)}{4} = 0.75 \text{ m,} \end{aligned}$$

$$\begin{aligned} y_{cm} &= \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3} \\ &= \frac{(0 + 0 + 4)}{4} = 1 \text{ m.} \end{aligned}$$



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Center-of-mass position, velocity, and acceleration (cont.)

For a system of a large number of particles with masses m (different particles need not have the same mass) moving on the xy -plane,

$$x_{cm} = \frac{1}{M} \sum mx, \quad y_{cm} = \frac{1}{M} \sum my \quad \text{where} \quad M = \sum m.$$

and,

$$v_{cm,x} = \frac{1}{M} \sum mv_x, \quad v_{cm,y} = \frac{1}{M} \sum mv_y, \quad P_x = Mv_{cm,x}, \quad P_y = Mv_{cm,y}.$$

$$a_{cm,x} = \frac{1}{M} \sum ma_x, \quad a_{cm,y} = \frac{1}{M} \sum ma_y, \quad F_x = Ma_{cm,x}, \quad F_y = Ma_{cm,y}.$$

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Center-of-mass position, velocity, and acceleration (cont.)

For a rigid body of mass M moving on the xy -plane,

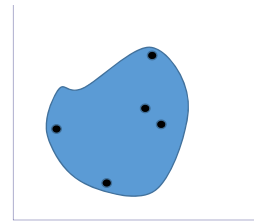
$$x_{cm} = \frac{1}{M} \int x dm, \quad y_{cm} = \frac{1}{M} \int y dm; \quad \text{where} \quad M = \int dm.$$

and,

$$v_{cm,x} = \frac{1}{M} \int v_x dm, \quad v_{cm,y} = \frac{1}{M} \int v_y dm, \quad P_x = Mv_{cm,x}, \quad P_y = Mv_{cm,y}.$$

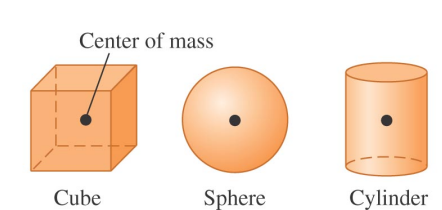
$$a_{cm,x} = \frac{1}{M} \int a_x dm, \quad a_{cm,y} = \frac{1}{M} \int a_y dm, \quad F_x = Ma_{cm,x}, \quad F_y = Ma_{cm,y}.$$

When a **rigid body** is acted on by **(external) forces**, the **center of mass** moves just as though all the mass were concentrated at that point and it were acted on by a net force equal to the sum of the **(external) forces** on the **rigid body**.

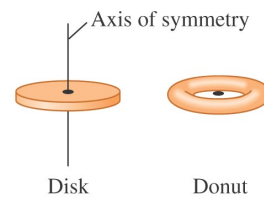


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Center-of-mass position, velocity, and acceleration (cont.)

Remark.

If a homogeneous object has a geometric center, that is where the center of mass is located.



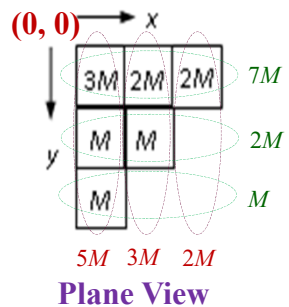
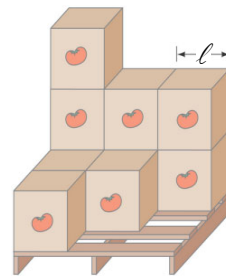
If an object has an axis of symmetry, the center of mass lies along it. As in the case of the donut, the center of mass may not be within the object.

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Question. A lightweight pallet has a load of identical boxes of tomato sauce, each of which is a cube of length ℓ . Find the center of gravity in the horizontal plane, so that the crane operator can pick up the load without tipping it.

Answer:

Let each box have a mass M . A top view of the pallet is shown, with the total mass of each stack listed.



Take the origin to be as shown in the figure.

Taking the moments w.r.t. $(0, 0)$,

$$x_{CM} = \frac{(5M)(0.5\ell) + (3M)(1.5\ell) + (2M)(2.5\ell)}{10M} = \boxed{1.2\ell}$$

$$y_{CM} = \frac{(7M)(0.5\ell) + (2M)(1.5\ell) + (1M)(2.5\ell)}{10M} = \boxed{0.9\ell}$$

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Question. A 45.0-kg woman stands up in a 60-kg canoe of 5.00 m long. She walks steadily from a point 1.00 m from one end to a point 1.00 m from the other end as shown in the Figure. If the resistance due to the motion of water is negligible, how far does the canoe move during this process?

Answer:

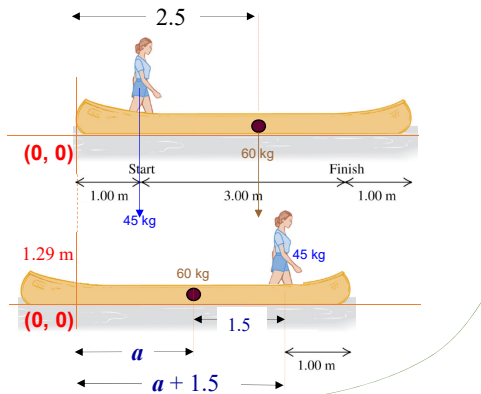
Let the super object contains the woman and the canoe. Let the Cartesian coordinate system be as shown on the figure. Initially the position of the center of mass of the super object is computed by moment as follows:

$$(45 \text{ kg} + 60 \text{ kg}) \times x_{\text{CM (start)}}^{\text{super object}} = 45 \text{ kg} \times 1 \text{ m} + 60 \text{ kg} \times 2.5 \text{ m}$$

Let a be the position of the center of the canoe when the woman is at the final position. The position of the woman on the same Cartesian coordinate system is $a + (2.5 - 1) = (a + 1.5) \text{ m}$. Similarly by taking the moment,

$$(45 \text{ kg} + 60 \text{ kg}) \times x_{\text{CM (finish)}}^{\text{super object}} = 45 \text{ kg} \times (a + 1.5) \text{ m} + 60 \text{ kg} \times a \text{ m}$$

In the absence of net external force, $x_{\text{CM (start)}}^{\text{super object}} = x_{\text{CM (finish)}}^{\text{super object}}$ w.r.t. the same $(0, 0)$.



$$45 \times 1 + 60 \times 2.5 = 45 \times (a + 1.5) + 60 \times a$$

$$195 = 45a + 67.5 + 60a$$

$$127.5 = 105a$$

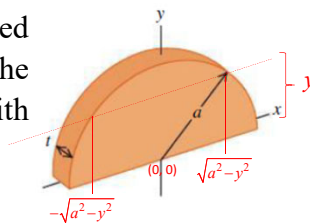
$$a = 1.214 \text{ m}$$

$$2.5 \text{ m} - 1.214 \text{ m} = 1.29 \text{ m.}$$

So the canoe moves left by 1.29 m.

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Problem. Using the coordinate system indicated below, calculate the x - and y -coordinates of the centre of mass of a semi-circular metal plate with uniform density ρ , radius a , and thickness t .



Answer:

Given a y where $0 \leq y \leq a$, the values of x will be $-\sqrt{a^2 - y^2} \leq x \leq \sqrt{a^2 - y^2}$. Let t be the thickness.

By symmetry, $x_{\text{cm}} = 0$. But more rigorously, $x_{\text{cm}} = \frac{1}{M} \int x dm = \frac{1}{M} \int_0^a \int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} x (\rho \times dV)$

$$= \frac{1}{M} \int_0^a \int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} x (\rho \times (dx \times dy \times t)) = \frac{\rho t}{M} \int_0^a \int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} x dx dy$$

$$\text{Since } \int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} x dx = \left[\frac{1}{2} x^2 \right]_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} = 0, \quad x_{\text{cm}} = \frac{\rho t}{M} \int_0^a \left(\int_{-\sqrt{a^2 - y^2}}^{\sqrt{a^2 - y^2}} x dx \right) dy$$

$$= \frac{\rho t}{M} \int_0^a 0 dy = 0$$

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$$y_{cm} = \frac{1}{M} \int y dm$$

$$= \frac{1}{M} \int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} y (\rho \times (dx \times dy \times t))$$

$$= \frac{\rho t}{M} \int_0^a \left(\int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} dx \right) y dy,$$

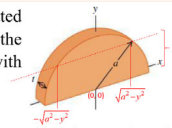
where $\int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} (1) dx = [x]_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}}$

$$= 2\sqrt{a^2-y^2}$$

$$\int_0^a (2\sqrt{a^2-y^2}) y dy = \int_0^a (-2y) \left(-\frac{2}{3} \right) \left(\frac{3}{2} \right) \sqrt{a^2-y^2} dy = -\frac{2}{3} \left[(a^2-y^2)^{\frac{3}{2}} \right]_0^a = \frac{2}{3} a^3$$

$$\rightarrow y_{cm} = \frac{\rho t}{M} \left(\frac{2}{3} a^3 \right) = \frac{\rho t}{\rho \times \left(\frac{\pi a^2}{2} \times t \right)} \left(\frac{2}{3} a^3 \right) = \frac{4}{3\pi} a \quad (\approx 0.4244a)$$

Problem. Using the coordinate system indicated below, calculate the x - and y -coordinates of the centre of mass of a semi-circular metal plate with uniform density ρ , radius a , and thickness t .



Answer:

Given a y where $0 \leq y \leq a$, the values of x will be $-\sqrt{a^2-y^2} \leq x \leq \sqrt{a^2-y^2}$. Let t be the thickness.

By symmetry, $x_{cm} = 0$. But more rigorously, $x_{cm} = \frac{1}{M} \int x dm = \frac{1}{M} \int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x (\rho \times dV)$

$$= \frac{1}{M} \int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x (\rho \times (dx \times dy \times t)) = \frac{\rho t}{M} \int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x dx dy$$

Since $\int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x dx = \left[\frac{1}{2} x^2 \right]_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} = 0$, $x_{cm} = \frac{\rho t}{M} \int_0^a \left(\int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x dx \right) dy$

$$= \frac{\rho t}{M} \int_0^a 0 dy = 0$$

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Example 3. The gravitational potential energy of a rigid body

Consider a rigid body of mass M freely falling from rest. For each particle of the rigid body with mass m freely falling through distance y ,

change in kinetic energy + change in gravitational potential energy = 0

$$\text{or } \left(\frac{1}{2} m v^2 - 0 \right) + (m g \times (-y) - m g \times 0) = 0 \rightarrow \frac{1}{2} m v^2 = m g y.$$

Here, $m g y$ is the gravitational potential energy of the particle with mass m . For the whole rigid body,

$$U_{\text{gravity}} = \sum m g y = g \sum m y = M g \frac{1}{M} \sum m y = M g \left(\frac{1}{M} \sum m y \right) = M g y_{cm}.$$

The gravitational potential energy of a rigid body is the same as though all the mass were concentrated at its centre of mass.

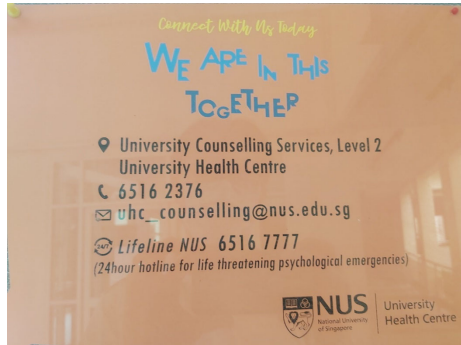
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Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019